Dynamics and Vibrations – Excercises

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Lecture 1: Introduction

26. August 2025

Exercise 1.1

We consider the rotation of a rigid body shown in **Figure 0.1** about the point O. The rod has mass m and the spring fastened to the rod at distance L from the point O has stiffness k. A force $F(t)$ is applied at a distance $L/2$ from the point O.

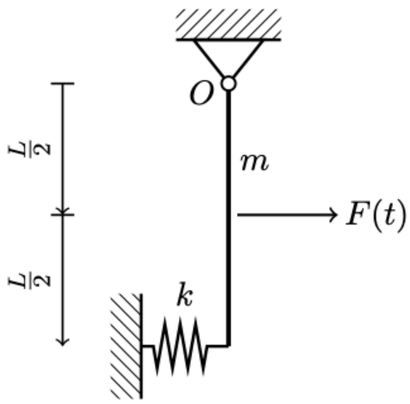


Figure 0.1:

a) Derive the non-linear equation of motion for the rotation of the rigid body.

We determine the resultant moment around O as:

$$I_O\ddot{\theta} = -kL\sin\theta L\cos\ \theta - mg\frac{L}{2}\sin\theta + F\frac{L}{2}\cos\theta$$

$$I_O\ddot{\theta} + \sin\theta\left(kL^2\cos\theta + mg\frac{L}{2}\right) = F\frac{L}{2}\cos\theta$$

.

The moment of inertia for a rod rotating about its end is given by $I_O = \frac{1}{3}mL^2$. Therefore we get:

$$\frac{1}{3}mL^2\ddot{\theta} + \sin\theta\left(kL^2\cos\theta + mg\frac{L}{2}\right) = F\frac{L}{2}\cos\theta.$$

b) Linearize the non-linear equation of motion using the assumption of small rotations.

By assuming $\cos\theta \approx 1$ and $\sin\theta \approx \theta$ we get:

$$\frac{1}{3}mL^2\ddot{\theta} + \theta\left(kL^2 + mg\frac{L}{2}\right) = F\frac{L}{2}.$$

Exercise 1.2

A person weighing 75 kg is standing on a scale in an elevator as shown on **Figure 0.2**. During the first 3 seconds of vertical motion upwards from rest, the force in the cable is 8300 N. Determine what the scale shows (in Newton) in this configuration. It is noted that the total mass of the elevator, person and scale is 750 kg.

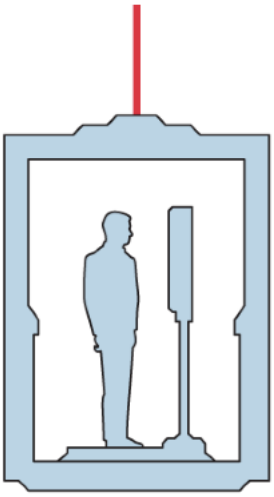


Figure 0.2:

Let $m_p = 75\,\text{kg}$ denote the mass of the person and $m = 750\,\text{kg}$ denote the entire mass of the system. Also let $F_c = 8300\,\text{N}$ denote the force in the cable. Newton's second law on the entire system gives

$$m\ddot{d} = F_c - mg \implies \ddot{d} = \frac{F_c - mg}{m}.$$

For the person inside the elevator using Newton's second law gives

$$m_p\ddot{d} = F_R - m_pg \implies F_R = m_p\ddot{d} + m_pg$$

where F_R is the reactant force the scale exerts on the person (i.e. the force the person exerts on the scale). Substituting in the previous expression for $\ddot{d}$ we get

$$
\begin{aligned}
F_R &= m_p\left(\ddot{d} + g\right) \\
&= m_p\left(\frac{F_c - mg}{m} + g\right) \\
&= m_p\left(\frac{F_c}{m}\right) \\
&= \frac{m_p F_c}{m} \\
&= \frac{75\,\mathrm{kg}\cdot 8300\,\mathrm{N}}{750\,\mathrm{kg}} \\
&= 830\,\mathrm{N}.
\end{aligned}
$$

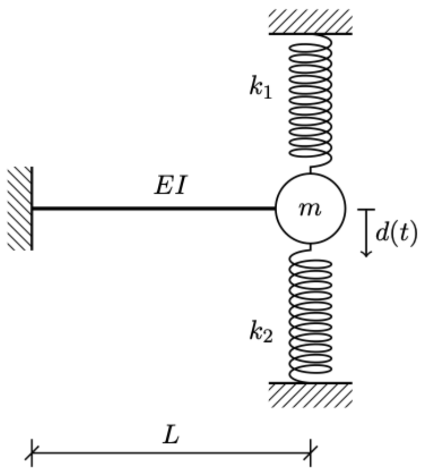
Therefore the scale will show a reading of 830 N.

Lecture 2: Free motion/eigenmotion of SDOF systems

2. September 2025

Exercise 2.1

We consider the shown system, which is to be analysed as an equivalent single-degree of freedom system (SDOF-system) with the transverse motion of the point mass m as the degree of freedom. It is noted, that the stiffness of the beam (without taking the discrete springs into account) against transverse direction in the free end due to a transverse force acting in this point is $k_{\text{beam}} = \frac{3EI}{L^3}$, where L, I, and E is the length, area-moment of inertia and the elastic modulus of the beam. Assume, that the beam and the discrete springs are mass-less.



a) Determine (symbolically) the undampened angular eigenfrequency ω_u of the equivalent SDOF-system.

The undampened angular eigenfrequency ω_u is given as:

$$\omega_u = \sqrt{\frac{k}{m}}.$$

In this case the three springs are connected in parallel and so:

$$k_{eq} = k_{\text{beam}} + k_1 + k_2 = k_1 + k_2 + \frac{3EI}{L^3}.$$

And therefore the undampened angular eigenfrequency of the system is

$$\omega_u = \sqrt{\frac{k_1 + k_2 + \frac{3EI}{L^3}}{m}}.$$

b) It is now noted that (in SI-units) $EI = 100$, $L = 1{,}3$, $m = 0{,}5$, $k_1 = k_2 = 25$, $d_0 = 0{,}1$ and $\dot{d}_0 = 0$. Determine the undampened eigenperiod of the equivalent SDOF-system.

Note that the connection between frequency as determined in a) and period is:

$$T = \frac{2\pi}{\omega_u} = \frac{2\pi}{\sqrt{\frac{2\cdot 25 + \frac{300}{1{,}3^3}}{0{,}5}}} = 0{,}33\,\text{s}.$$

c) Determine the maximal speed, that the equivalent SDOF-system experiences in its eigenmotion.

The displacement $d(t)$ is given as:

$$d(t) = A\cos\left(\omega_u t - \phi\right).$$

And the speed $\dot{d}(t)$ can be found simply vie differentiation as

$$\dot{d}(t) = -\omega_u A \sin\left(\omega_u t - \phi\right).$$

In its maximum this is

$$\max(\dot{d}(t)) = \omega_u A.$$

We also know that:

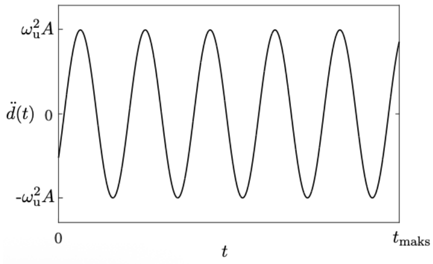
$$A = \sqrt{A_1^2 + A_2^2} = \sqrt{d_0^2 + \frac{\dot{d}_0^2}{\omega_u^2}} = d_0.$$

Therefore the maximum velocity is:

$$\max\left(\dot{d}(t)\right) = \omega_u A = \sqrt{\frac{k_1 + k_2 + \frac{3EI}{L^3}}{m}} \cdot d_0 = 1{,}73\,\frac{\text{m}}{\text{s}}.$$

Exercise 2.2

With reference to the following harmonic response choose the correct starting conditions.

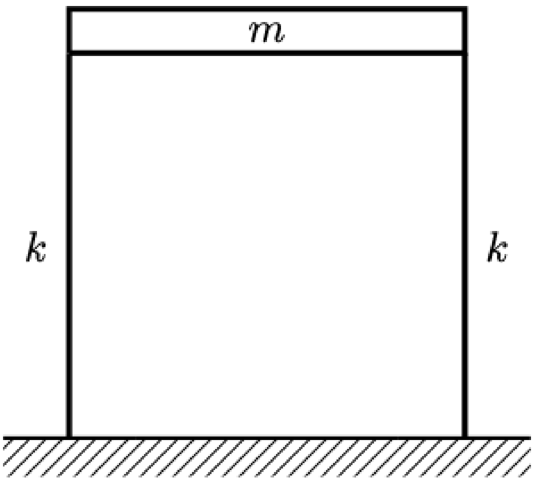


1. $d_0 < 0 \wedge \dot{d}_0 = 0$
2. $d_0 > 0 \wedge \dot{d}_0 = 0$
3. $d_0 < 0 \wedge \dot{d}_0 > 0$
4. $d_0 > 0 \wedge \dot{d}_0 < 0$
5. $d_0 = 0 \wedge \dot{d}_0 > 0$
6. $d_0 = 0 \wedge \dot{d}_0 < 0$

The plot looks to be showing sinusoidal acceleration. In the initial case the acceleration is negative, this means that the speed here is necessarily also negative, a negative speed means that the initial displacement must have been positive, hence the correct answer is 4

Exercise 2.3

We consider the shown system, which is to be analysed as an equivalent SDOF-system, wherein the degree of freedom is horizontal movement of the horizontal beam. It is noted that the horizontal beam has mass m and is assumed to be infinitely stiff, whilst the two vertical beams are assumed massless and with stiffness k.



a) Determine (symbolically) what the undampened angular eigenfrequency ω_u of the equivalent SDOF-system is.

The two beams are connected in parallel, hence:

$$k_{eq} = 2k \implies \omega_u = \sqrt{\frac{2k}{m}}.$$

b) It is noted that (in SI-units) $m = 3$, $k = 30$, $d_0 = -0{,}1$, and $\dot{d}_0 = 0{,}2$. Determine the initial acceleration of the equivalent SDOF-system

From Newton's second law we get:

$$m\ddot{d}(t) + kd(t) = 0 \implies m\ddot{d}_0 + k_{eq}d_0 = 0.$$

And now we can solve for $\ddot{d}_0$ as:

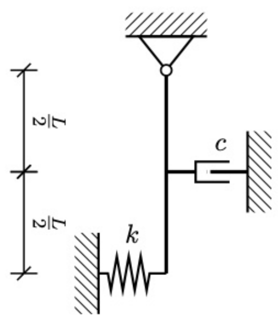
$$\ddot{d}_0 = \frac{-k_{eq}d_0}{m} = \frac{-60 \cdot (-0{,}1)}{3} = 2\,\frac{\mathrm{m}}{\mathrm{s}}.$$

Lecture 3: Damping of SDOF systems

9. September 2025

Exercise 3.1

We consider the shown system, which is to be analyzed as an equivalent SDOF system with the rotation of the rod $\theta(t)$ as the degree of freedom. It is assumed that the rod is uniform, infinitely stiff, has a mass of $m = 3\,\mathrm{kg}$, a length $L = 1{,}1\,\mathrm{m}$ and does not dissipate mechanical energy. A linear-viscous damping mechanism, which is producing subcritical damping with $\xi = 0{,}2$ and a spring with stiffness $k = 30\,\frac{\mathrm{N}}{\mathrm{m}}$ is applied as shown. The initial conditions are $\theta_0 = 0$ and $\dot{\theta} = 0{,}1$.



a) Derive the equation of motion for the equivalent SDOF system under the assumption of LTI conditions.

We imagine the rod being displaced a little to the right at the start since $d_0 > 0$. The force the spring will exert is:

$$F_s = -kL\sin\theta.$$

And the force that the damper will exert is

$$F_c = -c\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{L}{2}\sin\theta\right) = c\frac{L}{2}\dot{\theta}\cos\theta.$$

The force from gravity will be:

$$F_g = -mg\frac{L}{2}\sin\theta.$$

Now we take the moments about the support O as:

$$\begin{aligned}
I_O\ddot{\theta} &= F_s + F_c + F_g \\
&= -kL\sin\theta L\cos\theta - c\frac{L}{2}\dot{\theta}\cos\theta\frac{L}{2}\cos\theta - mg\frac{L}{2}\sin\theta \\
&= -kL^2\sin\theta\cos\theta - c\frac{L^2}{4}\dot{\theta}\cos^2\theta - mg\frac{L}{2}\sin\theta.
\end{aligned}$$

Under LTI assumptions we set $\sin\theta \approx \theta$ and $\cos\theta \approx 1$ and get:

$$\begin{aligned}
I_O\ddot{\theta} &= -kL^2\theta - c\frac{L^2}{4}\dot{\theta} - mg\frac{L}{2}\theta \\
&= -\theta\left(kL^2 + mg\frac{L}{2}\right) - c\frac{L^2}{4}\dot{\theta} \\
\implies I_O\ddot{\theta} + \frac{cL^2}{4}\dot{\theta} + \left(kL^2 + mg\frac{L}{2}\right)\theta &= 0.
\end{aligned}$$

Now inserting the fact that the moment of inertia of a rod rotating about its end is $I_O = \frac{1}{3}mL^2$ we get:

$$\frac{mL^2}{3}\ddot{\theta} + \frac{cL^2}{4}\dot{\theta} + L\left(kL + \frac{mg}{2}\right)\theta = 0.$$

b) Determine the undamped angular eigenfrequency ω_u.

For translation we normally have that

$$\begin{aligned}
m\ddot{d}(t) + c\dot{d}(t) + kd(t) &= 0 \\
\implies \ddot{d}(t) + \frac{c}{m}\dot{d}(t) + \frac{k}{m}d(t) &= 0 \\
\implies \ddot{d}(t) + 2\left(\omega_d\dot{d}(t) + \omega_u^2 d(t)\right) &= 0.
\end{aligned}$$

An analogous result is applicable for rotation, the eigenfrequency is:

$$\omega_u = \sqrt{\frac{k}{m}} \implies \omega_u = \sqrt{\frac{L\left(kL + \frac{mg}{2}\right)}{\frac{mL^2}{3}}}.$$

This can be simplified as:

$$\begin{aligned}
\omega_u &= \sqrt{\frac{3L\left(kL + \frac{mg}{2}\right)}{mL^2}} \\
&= \sqrt{\frac{3kL + \frac{3mg}{2}}{mL}}.
\end{aligned}$$

c) Solve the equation of motion in a) with the given initial conditions and sketch $\theta(t)$

We have the equation of motion:

$$\frac{mL^2}{3}\ddot{\theta} + \frac{cL^2}{4}\dot{\theta} + L\left(kL + \frac{mg}{2}\right)\theta = 0.$$

Now we set $m_{\text{sub}} \equiv \frac{mL^2}{3}$, $c_{\text{sub}} = \frac{cL^2}{4}$ and $k_{\text{sub}} = L\left(kL + \frac{mg}{2}\right)$. Then

$$m_{\text{sub}}\ddot{\theta} + c_{\text{sub}}\dot{\theta} + k_{\text{sub}}\theta = 0.$$

The solution of this for undercritial damping is:

$$d(t) = e^{-\xi\omega_u t}\left(A_1\cos(\omega_d t) + A_2\sin(\omega_d t)\right).$$

Where $\omega_d \equiv \omega_u\sqrt{1-\xi^2}$ and $\xi = \frac{c_{\text{sub}}}{2m_{\text{sub}}\omega_u}$. We can find an expression for ξ as

$$\begin{aligned}\xi &= \frac{c_{\text{sub}}}{2m_{\text{sub}}\omega_u}\\ &= \frac{\frac{cL^2}{4}}{\frac{2mL^2}{3}\omega_u}\\ &= \frac{3c}{8m\omega_u}.\end{aligned}$$

We have $\theta_0 = 0$, $\dot{\theta}_0 = 0{,}1$ so

$$\theta_0 = 0 = A_1.$$

As such we can find:

$$\begin{aligned}&\theta(t) = A_2 e^{-\xi\omega_u t}\sin(\omega_d t)\\ \implies\ &\dot{\theta}(t) = -\xi\omega_u A_2 e^{-\xi\omega_u t}\sin(\omega_d t) + \omega_d A_2 e^{-\xi\omega_u t}\cos(\omega_d t)\\ \implies\ &\dot{\theta}_0 = \omega_d A_2\\ \implies\ &A_2 = \frac{\dot{\theta}_0}{\omega_d}.\end{aligned}$$

Therefore the solution is:

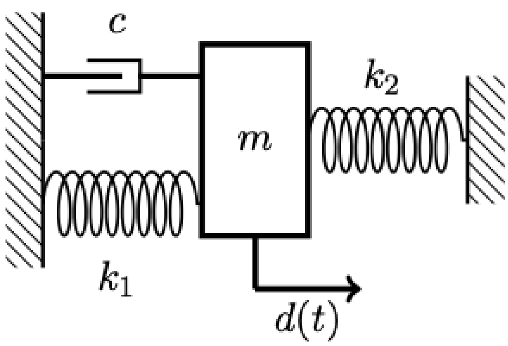
$$\theta(t) = \frac{\dot{\theta}_0}{\omega_d}e^{-\xi\omega_u t}\sin(\omega_d t).$$

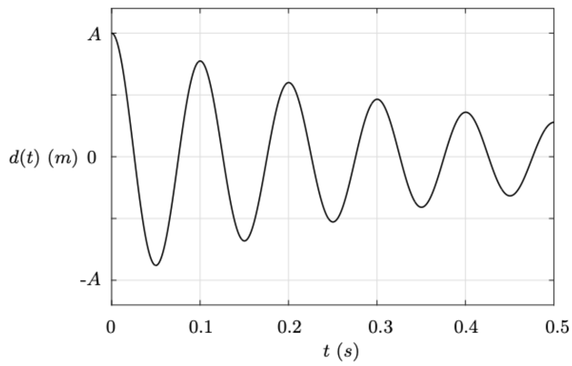
Exercise 3.2

We consider the shown mass-spring-damper system. Numerical values for the relevant physical parameters will be derived or provided during the exercise. I.e. avoid assuming any numerical values.

I an eigenmotion experiment with $d_0 > 0$ and $\dot{d}_0 = 0$ the movements shown underneath are created in the system. It is noted that the movement amplitude is reduced by 40% after two damped eigenperiods.

a) Determine the damping ratio ξ and determine if the system is undercritically damped, critically damped or overcritically damped.





The dampening ratio δ is defined as

$$\delta \equiv \frac{1}{j} \log\left(\frac{d(t)}{d(t+jT_d)}\right).$$

Where $j \in \mathbb{N}_{>0}$ is chosen such that $t + jT_d$ is the time j damped eigenperipds after t, where each eigenperiod is T_d. In this case we have:

$$\begin{aligned}\delta &= \frac{1}{2} \ln\left(\frac{d(t)}{\frac{3}{5}d(t)}\right) \\ &= \frac{1}{2} \ln \frac{5}{3} \\ &= 0{,}268.\end{aligned}$$

Hence:

$$\xi = \frac{\delta}{\sqrt{\delta^2 + 4\pi^2}} = \frac{0{,}268}{\sqrt{0{,}268^2 + 4\pi^2}} = 0{,}042.$$

Hence the system is undercritically damped.

b) Determine the damped angular eigenfrequency ω_d. (hint: look at the movement graph).

From the plot we see that $T_d = 0{,}1\,\mathrm{s}$, hence:

$$\omega_d = \frac{2\pi}{T_d} = 20\pi \mathrm{s}^{-1}.$$

c) Determine the undamped angular eigenfrequency ω_u.

Here we have:

$$\omega_u = \frac{\omega_d}{\sqrt{1-\xi^2}} = \frac{20\pi\mathrm{s}^{-1}}{\sqrt{1-0{,}042^2}} \approx 62{,}88\,\mathrm{s}^{-1}.$$

d) It is noted that $m = 2{,}2\,\mathrm{kg}$ and $k_1 = 3000\,\frac{\mathrm{N}}{\mathrm{m}}$. Determine k_2.

We know that:

$$k_{\mathrm{eq}} = k_1 + k_2$$

as the springs are connected in parallel. We have that:

$$\omega_u^2 = \frac{k_{\mathrm{eq}}}{m} = \frac{k_1 + k_2}{m} \implies k_2 = \omega_u^2 m - k_1.$$

Plugging in known values we get:

$$k_2 = \left(62{,}88\,\mathrm{s}^{-1}\right)^2 \cdot 2{,}2\,\mathrm{kg} - 3000\,\frac{\mathrm{N}}{\mathrm{m}} = 5{,}7\,\frac{\mathrm{kN}}{\mathrm{m}}.$$